Natural Language Generation

Question 1

Build and evaluate greedy and beam search algorithms from scratch (Tasks 1 and 2 in the script). In your report, include the output of of both greedy and beam search algorithms (hypotheses and log probabilities). You must follow the parameters specified in the script (prompt, beam size, max len). Describe the differences in the generated hypotheses and their log probabilities and discuss why they occur.

Greedy Search:

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Output:
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The cat slept on the floor, and the cat was still asleep.

"I was just trying to get her to sleep," she said. "I was trying to get her to sleep, but she was still asleep."

The cat

 Its log-probability:
----------------------------------------------------------------------------------------------------
-62.58576202392578
```

Beam Search:

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Output:
----------------------------------------------------------------------------------------------------
The cat slept on the floor of the house.

"It was like a nightmare," she said.

"It was like a nightmare. It was like a nightmare. It was like a nightmare. It was like a nightmare. It

 Its log-probability:
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-46.066965823905775
```

The beam search hypothesis achieved a higher log-probability than the greedy search hypothesis. This indicates that the beam search output is considered more probable by the language model overall and this is expected because greedy search selects only the single most probable token at each step without considering alternative continuations. In contrast, beam search keeps multiple candidate hypotheses at each decoding step and therefore it explores a broader search space.

The generated hypotheses also have some differences in style and structure. The greedy search output contains repetition around phrases such as *"trying to get her to sleep"* and *"still asleep"*. Beam search, on the other hand, has initially a good start but it also falls into a strong repetition loop with the phrase *"It was like a nightmare"*.

These differences occurs because the two algorithms make decisions differently. Greedy search always chooses the highest-probability next token. A token that appears best at the current step may not lead to the best overall continuation. Beam search reduces this problem by retaining the top scoring partial hypotheses and expanding them in parallel. This often improves the final sequence score. Beam search improves search quality over greedy search but it does not solve the repetition problem.

Question 2

Build and evaluate the beam search algorithm with a repetition penalty (Task 3). In your report, include the output of the beam search with the repetition penalty (hypothesis and log probability). You must follow the parameters specified in the script (prompt, beam size, max len, last tokens count). Describe the differences between the hypothesis generated in Task 3 to the hypothesis generated by the beam search in Task 2.

Beam Search with repeated penalty:

Output:

The cat slept on the floor of her room, and when she woke up to find that it had been eaten by a large black cat in the middle of her room, she immediately ran to help. She had no idea what was going on with the

Its log-probability:

-77.42537941224873

Beam search with repetition penalty discourage the decoder from reusing recently generated tokens which leads to less repetitive and more diverse sentence generation as compared to task 2.This improvement comes with the cost of a lower sequence probability in language models which can be seen through log probability values of beam search with penalty as compared to standard beam search.

Question 3

Experiment and assess sampling results in terms of coherence and diversity (Task 4).In your report, include the outputs of the four sampling methods for the specified prompt (random sampling, high temperature, low temperature, top-p sampling; implementations are already provided). Discuss the differences between the obtained hypotheses in terms of coherence and diversity.

Random Sampling output:

The cat slept on the floor of her room, while she held it to her chest.When Kiel went to check on him, a nurse found that Ejima had taken her cat.Kiel told the nurse that she

The random sampling is fairly diverse because it chooses random tokens from the distribution. The text is moderately coherent. The continuation of sentences are logically less connected.

High temperature output:

The cat slept on the walls. Two kids lay here — the boys went here with two very quiet girls and one with one pretty girl (for two months they hadn't got a boy since that day's story): one was not a parent and three

In high temperature scaling , the output is the most diverse but also least coherent.A high temperature flattens the probability which makes low probability words to be selected but as the output becomes more creative it also become less meaningless.

Low temperature output:

The cat slept on the floor, and the cat was still asleep.The cat was not the only one who had been affected.The cat was also found to have been in a coma.The cat had been found to have been in a coma

The output of this method is the most coherent because low temperature makes the probability distribution sharper and makes it like a greedy algorithm.The model chooses more probable tokenswhich comes at the cost of low diversity.

Top p output:

The cat slept on the door to his basement as a friend found it. That wasn't to say it didn't hurt, as the couple grew to love each other; in fact, while they probably already knew it had been broken, and that you.

Top-p sampling produces a balance between coherence and diversity.It avoids sampling from the full tail of ditributions but also allow some variety by selecting from a dynamic set of tokens.Although parts of the continuation are still odd but better than others.